$\big(\text{First num.} \neq \text{Second num.}\big) \times \big(\text{Third num.} \neq \text{First two num.}\big) \times \big(\text{Fourth num.} = \text{One of the num.}\big)$

$$P(X=4) = \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\frac{3}{n}$$

(i)

$$P(X=r) = \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{r-2}{n}\right)\frac{r-1}{n}$$

where $2 \leq r \leq n+1$

$\because X \in \{2,3,\ldots,n+1\}$, the sum of the probability $= 1$

$$\therefore \sum_{r=2}^{n+1} P(X=r) = 1$$

$$\therefore \frac{1}{n} + \left(1-\frac{1}{n}\right)\frac{2}{n} + \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\frac{3}{n} + \cdots = 1$$

(ii)

$$\boxed{\mathrm{E}(X) = \sum_{r=2}^{n+1} r\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{r-2}{n}\right)\frac{r-1}{n}}$$

(iii) $X \geq k \iff I_1, I_2, \ldots, I_{k-1}$ are distinct

$$\boxed{P(X \geq k) = \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-2}{n}\right)}$$

(iv)

$$\begin{aligned}
\mathrm{E}(Y) &= \sum_{r=1}^{N} rP(Y=r) \\
&= \sum_{r=1}^{N}\sum_{k=1}^{r} P(Y=r) \\
&= \sum_{k=1}^{N}\sum_{r=k}^{N} P(Y=r) \\
&= \sum_{k=1}^{N} P(Y \geq k)
\end{aligned}$$

Let $S_k = \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-1}{n}\right)$, where $S_1 = 1$
From part (iii) $\implies P(X \geq k+1) = S_k, \quad 1 \leq k \leq n$

$$\begin{aligned}
\mathrm{E}(X) &= \sum_{k=1}^{n+1} P(X \geq k) \\
&= P(X \geq 1) + \sum_{k=1}^{n} P(X \geq k+1) \\
&= 1 + \sum_{k=1}^{n} S_k
\end{aligned}$$

From part (ii) $\implies P(X = k+1) = S_k \frac{k}{n}, \quad 1 \leq k \leq n$
$\because \sum_{k=1}^{n} P(X = k+1) = 1 \quad \therefore \sum_{k=1}^{n} \frac{k}{n} S_k = 1$

$$\begin{aligned}
\mathrm{E}(X) &= \sum_{k=1}^{n} (k+1) P(X = k+1) \\
&= \sum_{k=1}^{n} (k+1) \frac{k}{n} S_k \\
&= \sum_{k=1}^{n} \frac{k^2}{n} S_k + \sum_{k=1}^{n} \frac{k}{n} S_k \\
&= 1 + \sum_{k=1}^{n} \frac{k^2}{n} S_k
\end{aligned}$$

$$1 + \sum_{k=1}^{n} S_k = 1 + \sum_{k=1}^{n} \frac{k^2}{n} S_k$$

$$\therefore \sum_{k=1}^{n} S_k \left(1 - \frac{k^2}{n}\right) = 0$$

$$\boxed{\left(1-\frac{1^2}{n}\right) + \left(1-\frac{1}{n}\right)\left(1-\frac{2^2}{n}\right) + \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\left(1-\frac{3^2}{n}\right) + \cdots = 0}$$

This solution was submitted by **Jon**
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